APPLIED LINGUISTICS DEPARTMENT

MIDTERM EXAMINATION

Prompt Engineering in Applied Linguistics

Practical Paper — Individual Project

TOTAL MARKS	DURATION
30	3 Hours (In-Lab)
PRIMARY AI TOOL	**EXTERNAL LLMS**
Antigravity	Permitted (label clearly)

SUBMISSION FORMAT

Single organized document: all prompts + outputs + verifications + final assembled product

Important Instructions — Read Carefully

1. **You are building a real, usable linguistics product** from scratch using AI.
2. Your paper must be **arranged in sections matching the clusters below** — in order.
3. For every prompt, you must write:
   `[PROMPT]` — what you typed into the AI | `[OUTPUT SUMMARY]` — key content received | `[VERIFY]` — how you checked the output is linguistically sound
4. If a cluster requires **external linguistic knowledge** (IPA, grammar theory, corpus data), consult another LLM. **Clearly label which LLM and for what purpose.**
5. Every cluster **connects to the next**. Earlier outputs must feed directly into later prompts.
6. At the end, assemble all components into **one final product document**.
7. **No cluster can be skipped.** Skipping a cluster breaks the product chain.

The Project

Build a Complete AI-Powered ESL Diagnostic and Teaching Toolkit for Pakistani University English Learners

You are a **prompt engineer and applied linguist** working for a Pakistani university's English Language Teaching (ELT) department. The department needs a complete, ready-to-use toolkit that:

→ Diagnoses where Pakistani university students struggle in English

→ Explains *why* they struggle using linguistic analysis

→ Provides lesson plans, exercises, and assessments to fix those struggles

→ Can be handed to any English teacher and used immediately

You will build this toolkit cluster by cluster using Antigravity. Every cluster produces one component of the final product. When all clusters are complete, you assemble them into the final toolkit.

End Product Structure

NO.	SECTION / COMPONENT
1	Learner Profile and Needs Analysis
2	Phonological Contrastive Analysis
3	Morphological Error Analysis
4	Syntactic Contrastive Analysis
5	Lexical and Collocational Analysis
6	Pragmatic Competence Gaps
7	Unified Error Taxonomy
8	Lesson Plan Bank (3 Lessons)
9	Exercise and Activity Bank
10	Assessment and Diagnostic Test
11	Verification and Quality Report
▶ Final Assembled Toolkit Document	

Cluster Overview

Cluster	Component	Section	Marks
1	Learner Profile and Needs Analysis	Section 1	3
2	Phonological Contrastive Analysis	Section 2	3
3	Morphological Error Analysis	Section 3	2
4	Syntactic Contrastive Analysis	Section 4	3
5	Lexical and Collocational Analysis	Section 5	2
6	Pragmatic Competence Gaps	Section 6	2
7	Unified Error Taxonomy	Section 7	3
8	Lesson Plan Bank	Section 8	4
9	Exercise and Activity Bank	Section 9	3
10	Assessment and Diagnostic Test	Section 10	3
11	Verification and Quality Report	Section 11	2
	TOTAL		30

01

CLUSTER 1 · SECTION 1 OF THE TOOLKIT

Learner Profile and Needs Analysis

3 Marks

Your Task: Before building anything, you must define who you are building for. A good linguistic toolkit is useless if it does not fit the learner.

1.1 Prompt

Prompt Antigravity to generate a detailed **Learner Profile** for Pakistani university students learning English as a second language. The profile must include: L1 background (Urdu/regional languages), educational context, English exposure history, approximate CEFR level range, and the most commonly reported challenges in reading, writing, speaking, and listening. Ask it to structure this as a **professional needs analysis document**.

1.2 Prompt

Based on the profile from Prompt 1.1, prompt Antigravity to identify the **top 6 linguistic problem areas** that the toolkit must address. Ask it to rank them by frequency and severity, and explain the linguistic reason behind each problem (e.g., *"article omission occurs because Urdu has no article system"*).

1.3 Prompt

Prompt Antigravity to generate a **one-page learner snapshot card** — a quick-reference document a teacher can read in 2 minutes. It should include: learner characteristics, key strengths, key gaps, and teaching implications.

⚙ VERIFY

Cross-check the 6 problem areas against at least one external source. Use another LLM or a linguistics resource to confirm these are documented issues for L1 Urdu learners. Label your source. Paste a 3-line confirmation note.

🔗 FEEDS INTO

> The 6 problem areas guide what gets analyzed in Clusters 2–6.

02

CLUSTER 2 · SECTION 2 OF THE TOOLKIT

Phonological Contrastive Analysis

3 Marks

Your Task: Conduct a systematic phonological comparison of Urdu and English to identify where pronunciation errors come from.

2.1 Prompt

Prompt Antigravity to conduct a **phonological contrastive analysis** between Urdu and English. Focus on: vowel inventory differences, consonants present in English but absent in Urdu, consonant cluster challenges, and syllable structure differences. Output should be formatted as a **comparison table**.

2.2 Prompt

Use another LLM or resource to retrieve the **IPA sound inventories** of both Urdu and English. Then return to Antigravity and prompt it to identify the **5 highest-interference phoneme pairs** — sounds Urdu speakers consistently substitute with an L1 equivalent. For each: English phoneme, Urdu substitute, and a minimal pair example.

2.3 Prompt

Prompt Antigravity to write a **"Phonological Profile of the Pakistani English Learner"** — a 300-word section for the toolkit that a non-specialist teacher can understand, explaining why their students sound the way they do and what to prioritize in pronunciation teaching.

⚙ VERIFY

Are the IPA symbols used correctly? Do the minimal pairs actually demonstrate the contrast? Prompt Antigravity to self-check the IPA table for errors. Paste its response. Correct any errors it finds.

🔗 FEEDS INTO

> Pronunciation errors identified here become phonology-related exercises in Cluster 9.

03

CLUSTER 3 · SECTION 3 OF THE TOOLKIT

Morphological Error Analysis

2 Marks

Your Task: Identify how Pakistani English learners struggle with word formation, inflection, and morphological rules.

3.1 Prompt

Prompt Antigravity to perform a **morphological contrastive analysis** between Urdu and English. Cover: plural formation, tense inflection, derivational morphology (prefixes/suffixes), and agreement morphology. Present findings as an annotated error table: *Error Type | Urdu Reason | English Rule | Example from student writing*.

3.2 Prompt

Prompt Antigravity to generate **15 authentic-looking morphological errors** that a Pakistani university student would realistically make in an essay. For each: show the incorrect form, the correct form, and classify the error type (omission, overgeneralization, transfer, etc.).

3.3 Prompt

Prompt Antigravity to write a **"Morphological Error Summary"** paragraph (150 words) for the toolkit — explaining the pattern of errors to teachers in plain language.

⚙ VERIFY

Review the 15 errors. Are they realistic? Flag any that seem unnatural for a Pakistani learner. Prompt Antigravity to review them and identify any that are "too advanced" or "unlikely." Document what you changed and why.

🔗 FEEDS INTO

These 15 errors become the basis of Cluster 9 morphology exercises.

04

CLUSTER 4 · SECTION 4 OF THE TOOLKIT

Syntactic Contrastive Analysis

3 Marks

Your Task: Map the structural differences between Urdu and English grammar that cause the most persistent syntactic errors.

4.1 Prompt

Prompt Antigravity to conduct a **syntactic contrastive analysis** between Urdu and English. Focus on: word order (SOV vs. SVO), verb placement, negation structure, question formation, relative clauses, and article use. Present as a structured comparison with English examples and Urdu-influenced error examples side by side.

4.2 Prompt

Prompt Antigravity to generate a **"Top 10 Syntactic Errors" list** specific to Pakistani university English learners. Each entry must include: the error pattern with an example sentence, the Urdu structural source, and the correct English reformulation.

4.3 Prompt

Research with another LLM: What does the **Contrastive Analysis Hypothesis (CAH)** predict about Urdu-English learning difficulty? Bring this back to Antigravity and prompt it to evaluate your Top 10 list against CAH predictions — which errors were predictable? Which were unexpected? Add this evaluation as a theoretical note in the toolkit section.

⚙ VERIFY

Take 3 errors from your Top 10 list. Write your own correct versions of each. Then prompt Antigravity to verify your corrections and explain the grammatical rule behind each. Paste both your versions and its verification.

🔗 FEEDS INTO

Top 10 syntactic errors directly structure the lesson plans in Cluster 8.

05

CLUSTER 5 · SECTION 5 OF THE TOOLKIT

Lexical and Collocational Analysis

2 Marks

Your Task: Identify how Pakistani learners misuse, under-use, or over-use vocabulary and collocations.

5.1 Prompt

Prompt Antigravity to identify **lexical and collocational errors** common in Pakistani university student writing. Focus on: false friends with Urdu, collocational mismatches (e.g., *"do a mistake"* instead of *"make a mistake"*), over-reliance on Urdu-borrowed formal register, and formulaic language gaps. Format as an error catalogue.

5.2 Prompt

Prompt Antigravity to create a **"Collocation Awareness" section** (20 entries) — a table showing: *Incorrect Collocation | Correct Collocation | Example sentence using the correct form*.

5.3 Prompt

Prompt Antigravity to generate a **"Key Academic Vocabulary" list of 30 words** that Pakistani university students need but often misuse or avoid. For each word include: the word, its meaning, a correct example sentence, and a common Pakistani learner mistake with it.

⚙ VERIFY

Pick 5 collocations from your table. Verify them using the British National Corpus (BNC) or another corpus resource (ask another LLM if needed). Are they real, high-frequency collocations? Correct any that are fabricated. Document your verification process.

🔗 FEEDS INTO

> Vocabulary list feeds into Cluster 9 exercises. Collocations feed into Cluster 10 diagnostic test.

06

CLUSTER 6 · SECTION 6 OF THE TOOLKIT

Pragmatic Competence Gaps

2 Marks

Your Task: Identify where Pakistani learners struggle not with grammar but with *how* language is used socially and academically.

6.1 Prompt

Prompt Antigravity to identify **5 key pragmatic competence gaps** in Pakistani university English learners. Cover: speech acts (requests, apologies, refusals), politeness strategies, academic writing register, turn-taking in discussions, and email/formal communication conventions. For each gap, explain the Urdu-English cultural-linguistic source.

6.2 Prompt

Prompt Antigravity to generate **10 sample student utterances** that are grammatically correct but pragmatically inappropriate in a Pakistani university context (e.g., an overly direct email to a professor). For each, provide the pragmatically improved version and explain the pragmatic principle applied.

6.3 Prompt

Prompt Antigravity to write a short **"Pragmatic Competence Guide for Students"** — a 200-word section written so a student can read it and understand why what they said was inappropriate and how to fix it.

⚙ VERIFY

Use another LLM to look up **Speech Act Theory** (Austin and Searle) and **Politeness Theory** (Brown and Levinson). Bring concise definitions back to Antigravity and prompt it to re-label your 10 utterances using correct theoretical terminology. This verifies the pragmatic analysis is theoretically grounded.

🔗 FEEDS INTO

> Pragmatic errors complete the full error picture before the unified taxonomy in Cluster 7.

07

CLUSTER 7 · SECTION 7 OF THE TOOLKIT — ANALYTICAL HEART

Unified Error Taxonomy

3 Marks

Your Task: This is the **analytical heart** of the toolkit. Synthesize everything from Clusters 2–6 into one organized, classified error taxonomy.

7.1 Prompt

Using your findings from Clusters 2–6, prompt Antigravity to build a **Unified Error Taxonomy Table**. The table must classify errors by: *Linguistic Level (Phonology / Morphology / Syntax / Lexis / Pragmatics) | Error Type | Example | Source (L1 Transfer / Overgeneralization / Avoidance) | Frequency (High / Medium / Low) | Teaching Priority (Critical / Important / Supplementary)*.

7.2 Prompt

Prompt Antigravity to apply **Corder's Error Analysis framework** (Interlingual vs. Intralingual errors) to your taxonomy. Ask it to re-classify each error and add a column to the table.

7.3 Prompt

Prompt Antigravity to write a **"Taxonomy Summary"** — a 250-word analytical interpretation of the table explaining what the errors reveal about Pakistani learners' interlanguage stage and what this means for teaching priorities.

VERIFY

Review the taxonomy table yourself. Are errors correctly categorized by linguistic level? Does the frequency rating match what Clusters 2–6 revealed? Prompt Antigravity to check for mis-categorized errors. Justify any disagreements. Document your reasoning in full.

FEEDS INTO

> The ranked "Critical" errors from this taxonomy directly determine what gets taught in Cluster 8.

08

CLUSTER 8 · SECTION 8 OF THE TOOLKIT

Lesson Plan Bank

4 Marks

Your Task: Design 3 ready-to-use, structured lesson plans targeting the top 3 "Critical" errors from your Cluster 7 taxonomy.

8.1 Prompt

From Cluster 7, identify your top 3 Critical Priority error types. Prompt Antigravity to design **Lesson Plan 1** addressing the highest-priority error. Structure: *Learning Objectives | Warm-Up Activity | Teaching Input (Explicit Explanation) | Guided Practice | Free Practice | Error Correction Strategy | Homework*. Duration: 50 minutes. Level: Pakistani university students.

8.2 Prompt

Repeat for **Lesson Plan 2** (2nd critical error). Prompt Antigravity to ensure this lesson explicitly references and builds on the previous lesson — include a review segment at the start.

8.3 Prompt

Repeat for **Lesson Plan 3** (3rd critical error). This lesson must include a **brief diagnostic check** at the start (5 minutes) to reveal whether students retained learning from Lessons 1 and 2.

8.4 Prompt

Prompt Antigravity to create a **Lesson Sequence Overview** — a one-page document showing how the 3 lessons connect, what each builds on, and how a teacher should pace them across one week.

⚙ VERIFY

Prompt Antigravity to check each lesson plan against this checklist: [1] Learning objectives are measurable (Bloom's Taxonomy verbs). [2] Activities match the linguistic

level of Pakistani university students. [3] Error correction strategies are pedagogically sound. [4] Lessons form a progression, not three isolated classes. Paste the checklist output and any revisions made.

🔗 **FEEDS INTO**

Lesson activities reference exercises from Cluster 9.

09

CLUSTER 9 · SECTION 9 OF THE TOOLKIT

Exercise and Activity Bank

3 Marks

Your Task: Build a bank of practice exercises covering all linguistic levels in the taxonomy. These exercises support Cluster 8 lessons and are given to teachers as standalone resources.

9.1 Prompt

Prompt Antigravity to generate **6 form-focused exercises** targeting morphological and syntactic errors (Clusters 3 and 4). Include: 2 gap-fill exercises, 2 error correction tasks, and 2 sentence transformation tasks. All sentences must use Pakistani-relevant content and vocabulary.

9.2 Prompt

Prompt Antigravity to generate **4 meaning-focused exercises** targeting lexis and collocations (Cluster 5). Include: 1 matching exercise, 1 collocation sorting task, 1 cloze passage with academic vocabulary, and 1 context-based word choice exercise.

9.3 Prompt

Prompt Antigravity to generate **3 pragmatic competence exercises** (Cluster 6). Include: 1 role-play scenario with guiding questions, 1 email rewriting task (inappropriate → appropriate), and 1 register analysis task (identify formal vs. informal features).

9.4 Prompt

Prompt Antigravity to create an **Answer Key** for all exercises, including brief explanations for commonly wrong answers.

⚙ VERIFY

Choose one exercise from each category and attempt it yourself as a student. Prompt Antigravity to check your answers against the key. Are the instructions clear? Is the

difficulty appropriate? Revise any ambiguous or misleveled exercise. Document what you changed.

🔗 **FEEDS INTO**

These exercises are referenced in Cluster 8 lesson plans and used as practice items in Cluster 10.

10

CLUSTER 10 · SECTION 10 OF THE TOOLKIT

Assessment and Diagnostic Test

3 Marks

Your Task: Create a diagnostic test for Day 1 to identify student weaknesses, and a summative test for after the lessons to measure progress.

10.1 Prompt

Prompt Antigravity to design a **30-minute Diagnostic Placement Test** covering all 5 linguistic levels from your taxonomy. Each section must have: 4 questions, clear instructions, a scoring key, and an interpretation guide (e.g., *"Score 0–2 in Syntax → Use Lesson Plan 2 + Exercise Set B"*).

10.2 Prompt

Prompt Antigravity to design a **Post-Teaching Assessment** (20 minutes) testing whether students improved in the 3 areas targeted by the Cluster 8 lesson plans. This test must be parallel in structure to the diagnostic test but with different items, so progress can be measured.

10.3 Prompt

Prompt Antigravity to create a **Score Interpretation and Action Guide** — a table showing teachers: if a student scores X in area Y, which lesson plan and which exercises from the toolkit to assign next.

⚙ VERIFY

Prompt Antigravity to perform a **content validity check**: Does the test actually test what it claims to test? Are question formats appropriate for the linguistic levels being tested? Is the post-teaching test parallel in difficulty to the diagnostic? Paste the check output. Revise any misaligned questions.

🔗 FEEDS INTO

Feeds directly into the Final Assembly as the toolkit's evaluation component.

11

CLUSTER 11 · SECTION 11 OF THE TOOLKIT — FINAL CHECK

Verification and Quality Report

2 Marks

Your Task: A professional product must be verified before it is published. Run a systematic quality check on the entire toolkit using Antigravity as your reviewer.

11.1 Prompt

Compile a summary of all 10 sections (1–2 sentences each). Prompt Antigravity to act as a **critical peer reviewer** — a senior applied linguist assessing the toolkit. Ask it to identify: 3 strengths, 3 weaknesses, and 3 missing elements that would make it more complete.

11.2 Prompt

Based on the weaknesses from Prompt 11.1, prompt Antigravity to **suggest specific revisions** to at least 2 sections. Implement ONE revision and document: what you changed, why, and how it improved the toolkit.

11.3 Prompt

Prompt Antigravity to write a final **"Quality Assurance Summary"** — a professional 200-word statement confirming that the toolkit: covers all major linguistic levels; is grounded in established linguistic theory (name at least 3 theories); is appropriate for the target learner profile; is practically implementable by a non-specialist teacher. This statement will appear as the foreword in the final toolkit.

⚙ VERIFY

This IS the verification cluster. No additional check needed — your documentation of what you changed in Prompt 11.2 is your evidence.

🔗 FEEDS INTO

All clusters complete. Proceed to Final Assembly.

Final Assembly — Complete Toolkit Document

Mandatory: An incomplete final assembly = an incomplete product and an incomplete grade.

PROMPT FINAL.1

Arrange all sections (1–11) into a single, organized document. Prompt Antigravity to write a **Professional Title Page** for the toolkit including: Title, Sub-title, Intended audience, Author (you), Date, and a 100-word abstract summarizing what the toolkit contains and how to use it.

PROMPT FINAL.2

Prompt Antigravity to write a **"How to Use This Toolkit"** guide — a 250-word practical introduction telling a teacher: where to start depending on their situation (new class, struggling students, specific skill focus), and how the sections connect to each other.

PROMPT FINAL.3

Prompt Antigravity to generate a **Table of Contents** for your assembled toolkit, with each section clearly labeled and a one-line description of what it contains.

Marking Rubric

Cluster	Component	Marks
1	Learner Profile and Needs Analysis	3
2	Phonological Contrastive Analysis	3
3	Morphological Error Analysis	2
4	Syntactic Contrastive Analysis	3
	TOTAL	**30**

Cluster	Component	Marks
5	Lexical and Collocational Analysis	2
6	Pragmatic Competence Gaps	2
7	Unified Error Taxonomy	3
8	Lesson Plan Bank	4
9	Exercise and Activity Bank	3
10	Assessment and Diagnostic Test	3
11	Verification and Quality Report	2
	TOTAL	**30**

What Markers Look For

Dimension	Strong Answer ✓	Weak Answer ✗
Prompt Quality	Specific, linguistically informed, builds on previous output	Vague, generic, same prompt copy-pasted throughout
Output Integration	Earlier outputs referenced and used in later prompts	Each cluster treated as isolated — no connections made
Verification	Student identifies errors, cross-checks, and revises	"It looks fine" — superficial or missing check
Linguistic Knowledge	Correct use of CAH, Error Analysis, Speech Acts, IPA	Theory names mentioned but clearly misunderstood
Product Coherence	Final toolkit reads as one unified, usable document	Sections feel disconnected; no logical flow
External LLM Use	Student knows WHEN to use another LLM and WHY	Never uses external sources, or uses them indiscriminately

Submission Checklist

- [ ] Paper is organized in 11 clearly labeled cluster sections, in order
- [ ] Every cluster has prompts labeled `[PROMPT X.X]`, `[OUTPUT SUMMARY]`, `[VERIFY]`
- [ ] External LLM usage is labeled with which LLM was used and for what purpose
- [ ] Final Assembly document is appended at the end
- [ ] All 11 toolkit sections are present in the final assembled product
- [ ] 150-word reflection included: What was the hardest prompt to write, and why?